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import pandas as pd
from azure.core.credentials import AzureKeyCredential
from azure.ai.textanalytics import TextAnalyticsClient

# ===== Azure credentials =====
key =
"8FK3CXRUDdpUp4uHnnzGErUKBXQuKA2fToeR1Dv6Axxoi8yHB7xfJQQJ99CCACYeBjFXJ3w3AAAAACOGqN
sZ"
endpoint = "https://federer-sentiment-language.cognitiveservices.azure.com/"

# ===== Create client =====
credential = AzureKeyCredential(key)
client = TextAnalyticsClient(endpoint=endpoint, credential=credential)

# ===== Read CSV =====
df = pd.read_csv("youtube_comments.csv")
documents = df["text"].astype(str).tolist()

# ===== Run sentiment analysis in batches of 10 =====
sentiments = []
positive_scores = []
neutral_scores = []
negative_scores = []

batch_size = 10

for i in range(0, len(documents), batch_size):
    batch = documents[i:i + batch_size]
    results = client.analyze_sentiment(batch)

    for result in results:
        if not result.is_error:
            sentiments.append(result.sentiment)
            positive_scores.append(result.confidence_scores.positive)
            neutral_scores.append(result.confidence_scores.neutral)
            negative_scores.append(result.confidence_scores.negative)
        else:
            sentiments.append("error")
            positive_scores.append(None)
            neutral_scores.append(None)
            negative_scores.append(None)

# ===== Save output =====
df["sentiment"] = sentiments
df["positive_score"] = positive_scores
df["neutral_score"] = neutral_scores
df["negative_score"] = negative_scores

df.to_csv("sentiment_results.csv", index=False)

print("Sentiment analysis complete. Results saved to sentiment_results.csv")

from azure.storage.blob import BlobServiceClient

# ===== Azure Blob Storage =====
connection_string =
"DefaultEndpointsProtocol=https;AccountName=federersentimentstorage;AccountKey=JRXi
G/ykZyEvaX7sdCd8YKf/
3FB7BXqycfiJU0w5pllTVj508j5u6x2TdAM203onIWX5lDj0dEKe+ASTyzY0XQ==;EndpointSuffix=cor

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e.windows.net"
container_name = "results"
blob_name = "sentiment_results_from_python.csv"

blob_service_client = BlobServiceClient.from_connection_string(connection_string)
blob_client = blob_service_client.get_blob_client(container=container_name,
blob=blob_name)

with open("sentiment_results.csv", "rb") as data:
    blob_client.upload_blob(data, overwrite=True)

print("File uploaded to Azure Blob Storage successfully.")
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